PROBLEM 1.6

CSTR Mass Balance • Washout • Integrating Factor

Saline Mixing Tank Dynamics (CSTR)

Original statement · ProblemsCh1.pdf, p. 2

PHASE 1: PROBLEM STATEMENT & HYPOTHESES

Problem statement: A tank contains 100 liters of water with 1 kg of salt dissolved in it (uniformly). At time $t_0 = 0$, water without salt flows into the tank at a rate of 1 liter per minute, while the mixture flows out at the same rate.

(i) Write a differential equation for $x(t)$, which is the number of kg of salt per litre of water dissolved at time $t > 0$.
(ii) How much salt is dissolved in the tank as $t \to \infty$?
(iii) What happens if water with s kg of salt per liter flows into the tank at a given rate while the mixture (that is kept uniform) flows out at the same rate?

PHASE 2: FUNDAMENTAL FORMULATION

As in the official statement, $x(t)$ is the salt **concentration** (kg of salt per litre, units $\mathrm{kg/L}$). The total dissolved salt mass is $M(t) = V(t)\,x(t)$ (units kg). Species conservation of dissolved salt mass: $\frac{dM}{dt} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}} = Q_{\text{in}} C_{\text{in}} - Q_{\text{out}}\,x(t)$. Since $Q_{\text{in}} = Q_{\text{out}} = Q, V(t) \equiv V_0 = 100\ \mathrm{L}$, so $M(t) = 100\,x(t)$ and the initial concentration is $x(0) = \frac{1\ \mathrm{kg}}{100\ \mathrm{L}} = 0.01\ \mathrm{kg/L}$.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): Pure Water Flushing ($C_{\text{in}} = 0$):

$$\dot{m}_{\text{in}} = 1 \times 0 = 0\ \mathrm{kg/min}, \quad \dot{m}_{\text{out}} = 1 \times x(t) = x(t)\ \mathrm{kg/min}$$

With $M = 100\,x$, the balance $\frac{dM}{dt} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}}$ becomes $100\,\frac{dx}{dt} = -x(t)$, i.e.

$$\boxed{\frac{\mathbf{dx}}{\mathbf{dt}} = -\frac{\mathbf{x(t)}}{\mathbf{100}}, \quad x(0) = 0.01\ \mathrm{kg/L}} \qquad (1.6.1)$$

Part (ii): Analytical Solution and Limit as $t \to \infty$:

$$\int_{0.01}^{x(t)} \frac{d\xi}{\xi} = \int_0^t -\frac{1}{100} ds \implies \ln \frac{x(t)}{0.01} = -\frac{t}{100} \implies \mathbf{x(t) = 0.01\, e^{-t/100}\ \text{kg/L}}$$

The dissolved salt mass is $M(t) = 100\, x(t) = e^{-t/100}$ kg. Taking the limit as $t \to \infty$:

$$\lim_{t\to\infty} x(t) = \mathbf{0}\ \text{kg/L}, \qquad \lim_{t\to\infty} M(t) = \lim_{t\to\infty} e^{-t/100} = \mathbf{0}\ \text{kg}$$

Part (iii): Saline Inflow at Concentration s kg/L:

Now $\dot{m}_{\text{in}} = Q \cdot s = 1 \cdot s = s$ kg/min, so $100\, \frac{dx}{dt} = s - x(t)$, leading to the inhomogeneous linear ODE:

$$\frac{dx}{dt} + \frac{1}{100} x(t) = \frac{s}{100} \tag{1.6.2}$$

Multiplying by integrating factor $\mu(t) = \exp\left(\int \frac{1}{100} dt\right) = e^{t/100}$:

$$\frac{d}{dt}\left[x(t) e^{t/100}\right] = \frac{s}{100} e^{t/100}$$

Integrating both sides from 0 to t (Barrow's rule):

$$x(t)e^{t/100} - x(0) = \frac{s}{100}\int_0^t e^{\tau/100} d\tau = s\left[e^{t/100} - 1\right]$$

With $x(0) = 0.01$ kg/L:

$$x(t)e^{t/100} = 0.01 + s e^{t/100} - s = s e^{t/100} + (0.01 - s)$$

Dividing by $e^{t/100}$:

$$\mathbf{x(t) = s + (0.01 - s) e^{-t/100}\ \text{kg/L}} \tag{1.6.3}$$

The corresponding dissolved salt mass is $M(t) = 100\, x(t) = 100s + (1 - 100s)e^{-t/100}$ kg. As $t \to \infty$, $e^{-t/100} \to 0$:

$$\lim_{t\to\infty} x(t) = \mathbf{s}\ \text{kg/L}, \qquad \lim_{t\to\infty} M(t) = \mathbf{100s}\ \text{kg}$$

PHASE 4: FINAL RESULT & ANALYSIS

BOXED RESULT

$$\text{Washout: } x(t) = 0.01\, e^{-t/100}\ \text{kg/L},\ M(t) = e^{-t/100}\ \text{kg},\ \lim_{t\to\infty} M(t) = 0$$

$$\text{Saline Inflow: } x(t) = s + (0.01 - s)e^{-t/100}\ \text{kg/L},\ \lim_{t\to\infty} x(t) = s\ \text{kg/L},\ \lim_{t\to\infty} M(t) = 100s\ \text{kg}$$

Physical Interpretation: The asymptotic equilibrium concentration is the inlet concentration s kg/L, i.e. a dissolved mass $100s$ kg (tank volume 100 L times s kg/L).